



Q1. In my counter circuit using TSP1738 IR sensor and CD4026 decade counter, the counting starts automatically even if clock input pin of CD4026 is not connected. When TSP1738 is connected to clock pin, it does not count properly. What could be the reason?

Kota Sharan Pai

A1. There can be several reasons for this misbehavior of the circuit. First and main reason could be the floating condition at the clock input pin of CD4026 counter. The best solution is to connect a pull-down resistor, say 10-kilo-ohm, at clock input pin. Otherwise, the digital display will automatically start counting or create problem randomly.

Second, if it does not count properly after TSP1738 is connected to clock pin, it could be due to false triggering of the infrared (IR) sensor. TSP1738 is sensitive and has inherent false triggering problem if exposed to light. Make sure that your IR sensor is not exposed to any light, and that there is

no electrical sparking near your circuit. You may use a 470µF electrolytic capacitor across pins 1 and 3 of TSP1738. You can also cover the sensor in a suitable enclosure with a hole in front to receive IR beam from the transmitter.

The third reason could be frequency mismatch between the transmitter and the receiver. Make sure that the transmitter frequency is tuned to around 38kHz because TSP1738 is designed to be operated at 38kHz frequency. If the problem persists, try replacing TSP1738 with a fresh one.

Another reason could be loose connections, either in the receiver or the transmitter circuit.

Q2. The passive keyless entry system in my car was working fine till recently, but now the response time has decreased. When I touch the antenna it recognises the key and unlocks the doors. What could have gone wrong?

Sabih

A2. One of the possible reasons could be a weak battery or lose connections in power supply.

Another could be misalignment of sensitivity settings, or defective proximity sensors in the system. The sensor is used to detect the presence or absence of an object, or driver in this case, using electromagnetic field, light, and sound. There are

many proximity sensors including capacitive, inductive, photoelectric, ultrasonic, which are meant for specific applications and environments. In most cases, capacitive proximity sensors are used in passive keyless entry systems.

The best solution in such cases is to contact the car dealer, or manufacturer of the device, and get it checked.

Q3. What is LoRa and what is LoRaWAN? Which LoRaWAN modules are easily available in the market?

Pamarthi Kanakaraja

A2. LoRa stands for Long Range. LoRaWAN, also known as LPWAN, stands for Long Range Wide Area Network. LoRaWAN is a protocol designed to connect devices to the Internet for IoT applications. The LoRa physical layer in the network enables the long-range communication link and LoRaWAN defines the communication protocol and system architecture for the network. LoRaWAN is recommended for long-range device communications. Some popular LoRa modules easily available in the market include CMWX1ZZABZ, SX1276RF1JAS, SX1276, RHF0M301, iM980A, and RHF76-052.



Typical LoRaWAN module
(Credit: www.cooking-hacks.com)